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RBE211TC DYNAMIC SYSTEMS

VIBRATION OF MECHANICAL SYSTEMS – SDOF SYSTEMS (PART 1)

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Lecture Outline

1. The link between second-order response and vibration
2. Free vibration versus forced vibration
3. Harmonic excitation of SDOF systems
4. Transient and steady-state response
5. Amplitude ratio and resonance

From Dynamic System Response to Vibration

Over the past few weeks, we have:

- developed mathematical models for dynamic systems
- studied first- and second-order system response
- observed that second-order systems may oscillate

When this oscillatory behaviour occurs in mechanical systems, we call it **vibration**.

Why Study Vibration in Robotics?

Vibration can influence:

- motion accuracy
- structural integrity
- control performance
- noise and wear
- safety and reliability

Understanding vibration helps engineers:

- predict system behaviour
- avoid resonance
- improve design and damping

Mechanical Elements Behind Vibration

A mechanical dynamic system is typically composed of:

- **Inertia (mass)** — stores kinetic energy
- **Stiffness (spring)** — stores potential energy
- **Damping (damper)** — dissipates energy

These elements determine how a system vibrates.

Recap: Standard Form of an SDOF Mechanical System

For a mass-spring-damper system:

$$m\ddot{x} + c\dot{x} + kx = 0$$

Normalized form:

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0$$

where:

- ω_n: natural frequency
- ζ: damping ratio

Free Vibration Behaviour

Depending on damping ratio:

- **Underdamped**: oscillatory response with decaying amplitude
- **Critically damped**: fastest non-oscillatory return
- **Overdamped**: non-oscillatory but slower return

Key idea:

- free vibration is caused by initial conditions only

From Free Vibration to Forced Vibration

Free vibration

- motion caused by initial displacement or initial velocity
- no continuous external input

Forced vibration

- motion caused by an external time-varying input
- excitation continues during the motion

Typical forcing:

$$F(t) = F_0 \sin(\omega t)$$

Harmonic Excitation

A harmonic force can be written as:

$$F(t) = F_0 \sin(\omega t)$$

where:

- F_0: force amplitude
- ω: excitation frequency

Examples in engineering:

- rotating imbalance
- periodic motor torque
- gear meshing
- repetitive loading in robotic systems

Equation of Motion Under Harmonic Excitation

For an SDOF system subjected to a harmonic force:

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin(\omega t)$$

Key observation:

- the response now depends on both
  - the system parameters (m, c, k)
  - the excitation frequency ω

Transient and Steady-State Response

For a harmonically excited system, the total response consists of:

- a **transient response**
- a **steady-state response**

So,

$$x(t) = x_t(t) + x_s(t)$$

Key idea:

- transient part depends on initial conditions
- steady-state part is sustained by the external force

Why Is Steady-State Response Important?

In many practical situations:

- initial effects disappear after some time
- long-term motion is dominated by the forced response

Therefore, engineers are often interested in:

- response amplitude
- phase relationship
- effect of excitation frequency

Undamped Forced Vibration

To understand the basic mechanism clearly, first consider the undamped case:

$$m\ddot{x} + kx = F_0 \sin(\omega t)$$

Assume the steady-state response takes the form:

$$x_s(t) = X \sin(\omega t)$$

where X is the response amplitude.

Steady-State Amplitude of the Undamped System

For the undamped system, the steady-state amplitude is

$$X = \frac{F_0/k}{1 - (\omega/\omega_n)^2}$$

where

$$\omega_n = \sqrt{\frac{k}{m}}$$

and

$$\frac{F_0}{k}$$

is the static deflection under the force amplitude F_0.

Frequency Ratio and Magnification Factor

Define the frequency ratio

$$r = \frac{\omega}{\omega_n}$$

Then the amplitude ratio becomes

$$\frac{X}{F_0/k} = \frac{1}{1 - r^2}$$

This ratio is called the **magnification factor** or **amplitude ratio**.

Interpretation of the Frequency Ratio

Three important cases:

- $r < 1$
  excitation frequency is below natural frequency
- $r = 1$
  excitation frequency equals natural frequency
- $r > 1$
  excitation frequency is above natural frequency

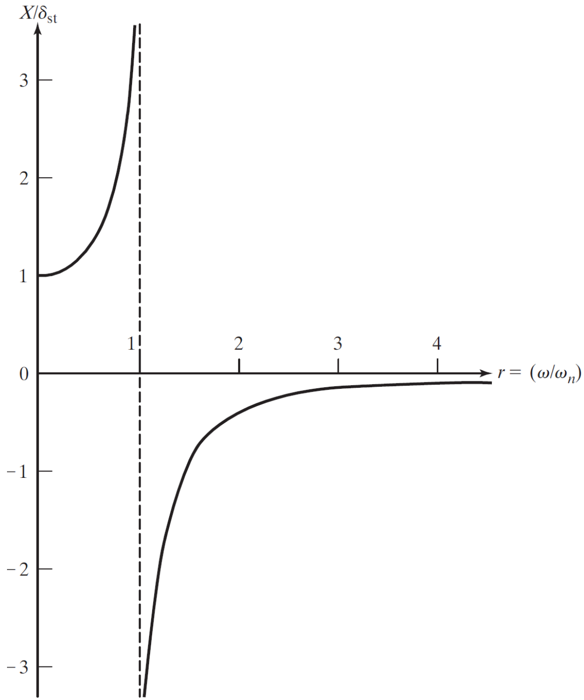
Key question:
How does the response change as r varies?

Behaviour for Different Frequency Ratios

Amplitude ratio:

$$\frac{X}{F_0/k} = \frac{1}{1 - r^2}$$

- **Below resonance** $(r < 1)$
  response amplitude increases as r increases
- **At resonance** $(r = 1)$
  response becomes very large
- **Above resonance** $(r > 1)$
  response amplitude decreases



Physical idea:
The response is most severe when excitation frequency is close to natural frequency.

Resonance in an Undamped System

When

$$\omega = \omega_n$$

the denominator in the amplitude expression becomes zero.

Implication:

- the ideal undamped model predicts unbounded response
- energy is added at the most effective frequency
- resonance may cause severe vibration

Undamped Forced Response

Example 1

A mass-spring system has:

- $m = 2\ kg$
- $k = 200\ N/m$
- $F_0 = 10\ N$

It is subjected to the harmonic force

$$F(t) = 10\sin(8t)$$

Determine:

- the natural frequency
- the frequency ratio
- the steady-state amplitude
- whether the system operates close to resonance

Undamped Forced Response

Example 1

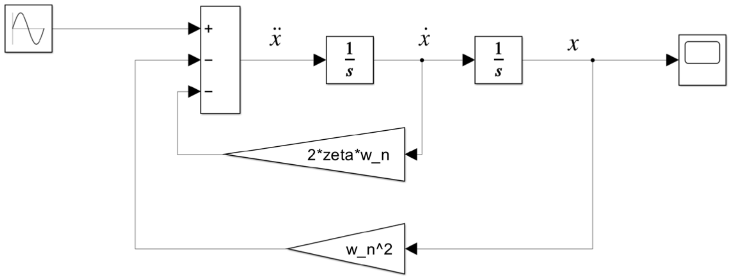
Undamped Forced Response

Example 1

Simulation Check: Effect of Excitation Frequency

Compare the response for:

- $r = 0.5$
- $r = 0.95$
- $r = 1.5$



Simulink model of a second-order system.

Observe:

- response amplitude
- oscillation pattern
- severity near resonance

Key question:
How does the time response change as the excitation frequency moves relative to the natural frequency?

Forced Vibration of a Damped SDOF System

Now consider the damped system:

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin(\omega t)$$

Assume the steady-state response takes the form:

$$x_s(t) = X \sin(\omega t - \phi)$$

where:

- X: response amplitude
- ϕ: phase lag

Steady-State Amplitude of the Damped System

For the damped system, the amplitude ratio is

$$\frac{X}{F_0/k} = \frac{1}{\sqrt{\left(1 - r^2\right)^2 + \left(2\zeta r\right)^2}}$$

where:

- $r = \dfrac{\omega}{\omega_n}$
- ζ: damping ratio

Key idea:

- damping limits the response amplitude

Phase Lag in Forced Vibration

The phase lag is given by

$$\tan \phi = \frac{2\zeta r}{1 - r^2}$$

Interpretation:

- the response does not occur exactly in phase with the excitation
- phase lag depends on frequency ratio and damping ratio

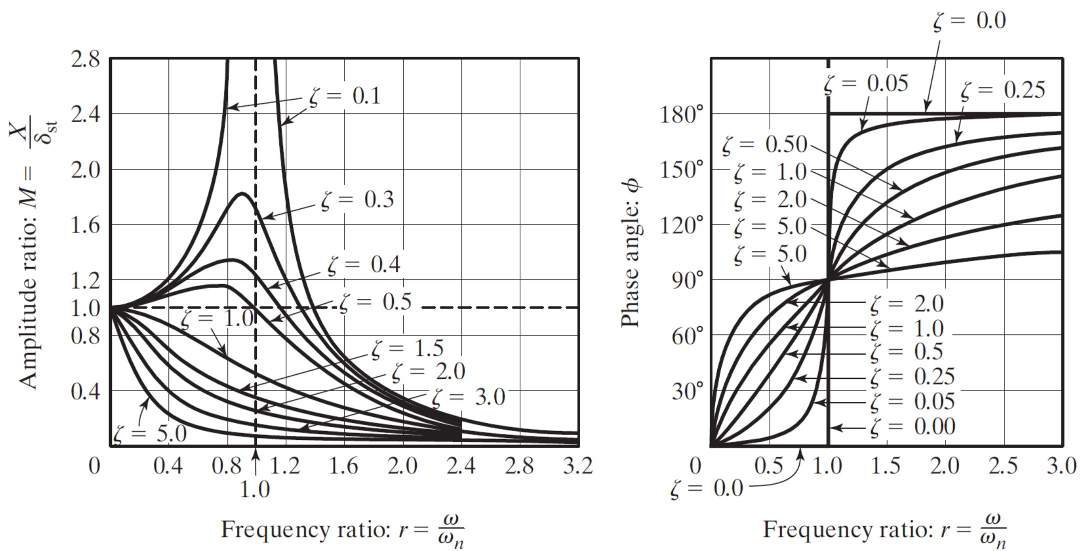
Effect of Damping on the Resonance Peak

Amplitude ratio:

$$\frac{X}{F_0/k} = \frac{1}{\sqrt{\left(1 - r^2\right)^2 + \left(2\zeta r\right)^2}}$$

Phase lag:

$$\tan\phi = \frac{2\zeta r}{1 - r^2}$$

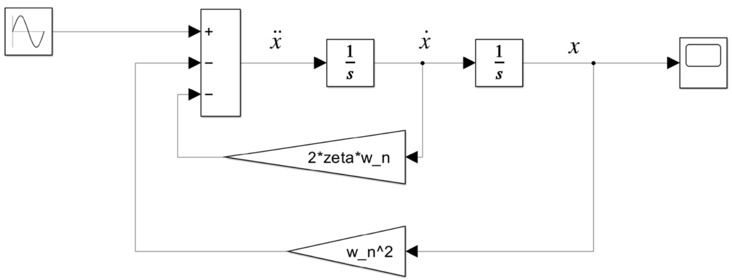


Variation of $\frac{X}{F_0/k}$ and ϕ with frequency ratio r.

Simulation Check: Effect of Damping Ratio

Compare the response for:

- $\zeta = 0.02$
- $\zeta = 0.10$
- $\zeta = 0.30$



Simulink model of a second-order system.

Observe:

- response amplitude
- severity of oscillation
- influence of damping near resonance

Key question:
How does the time response change as the damping ratio varies?

Example 2

A damped SDOF system has:

- $m = 1\,kg$
- $k = 100\,N/m$
- $\zeta = 0.2$
- $F_0 = 5\,N$

The excitation frequency is

$$\omega = 10\,rad/s$$

Determine:

- the natural frequency
- the frequency ratio
- the amplitude ratio
- the steady-state amplitude

Example 2

Example 2

Comparison of Operating Conditions

Example 3

For the same damped system, compare the following cases:

- **Case A:** $r = 0.6,\ \zeta = 0.05$
- **Case B:** $r = 1.0,\ \zeta = 0.20$
- **Case C:** $r = 1.4,\ \zeta = 0.05$

Discuss:

- which case is closest to resonance
- which case is likely to produce the largest response
- how damping affects the outcome

Comparison of Operating Conditions

Example 3

Review

1. Forced vibration of SDOF systems
2. Harmonic excitation
3. Transient and steady-state response
4. Amplitude ratio and frequency ratio
5. Resonance
6. Effect of damping on amplitude and phase